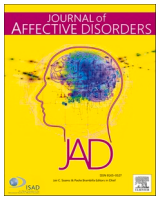
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Research paper



Development and multi-center validation of a school-home integrated machine learning model for early screening of attention-deficit/hyperactivity disorder in school-aged children

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ABSTRACT

Background: Attention-deficit/hyperactivity disorder (ADHD) is frequently under-identified in community settings, delaying necessary intervention. This study aimed to develop and validate a multidimensional Early Screening Prediction Model (ESPM-ADHD) for children aged 6 to 18 years, integrating school-home collaborative data.

Methods: A large-scale cross-sectional survey was conducted from January to December 2023 across three distinct regions in China. The dataset was divided into a development cohort (Chongqing, n = 15,085) and two independent external validation cohorts (Shaanxi, n = 7435; Yunnan, n = 4206). Machine learning algorithms, including Random Forest, XGBoost, LightGBM, and Logistic Regression, were trained using 15 selected features. The best-performing model was deployed as a web-based tool.

Results: A total of 26,726 valid questionnaires were analyzed, with an overall ADHD risk prevalence of roughly 2%. Among the tested algorithms, the Random Forest model demonstrated the most robust performance. In external validation, the model maintained high discriminatory power across diverse populations. Specifically, it achieved an AUPRC of 0.298 and an AUROC of 0.929 in the Shaanxi cohort, and an AUPRC of 0.253 and an AUROC of 0.907 in the Yunnan cohort. SHAP analysis identified antisocial behavior, academic performance (native language and math), and daytime sleepiness as the top predictors.

Conclusions: The ESPM-ADHD, constructed using the Random Forest algorithm, effectively identifies children at risk for ADHD by leveraging easily assessable demographic, academic, and somatic indicators. The developed online calculator provides a practical, objective triage tool for clinicians and guardians to facilitate timely referral and diagnosis.

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) remains one of the most prevalent neurodevelopmental disorders globally, characterized by a persistent pattern of inattention, hyperactivity, and impulsivity (Faraone et al., 2024). According to recent meta-analyses, the global prevalence of ADHD in children and adolescents is estimated at approximately 8.0% (Ayano et al., 2023), with data from China indicating a comparable prevalence of 6.4% among school-aged children (Li et al., 2022). The disorder imposes a substantial burden, with approximately two-thirds of cases persisting into adulthood (Faraone et al., 2021), leading to long-term adverse outcomes in academic achievement, social functioning, and mental health (Jennum et al., 2025).

Despite its high prevalence, the early identification of ADHD remains a significant clinical challenge. Current identification strategies depend heavily on subjective reporting and behavioral observations, which are susceptible to observer bias and variability. This limitation disproportionately affects children with mild symptomatology or the

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predominantly inattentive presentation, whose conditions are often under-identified until significant functional impairment occurs (Young et al., 2020b). Consequently, there is a critical need for objective, accessible screening tools capable of leveraging readily available data to effectively triage at-risk children for further clinical evaluation.

The development of an effective screening tool necessitates the identification of robust predictors that extend beyond core symptomatology. Emerging evidence suggests that ADHD is rarely an isolated condition; rather, it manifests with a constellation of demographic traits and comorbidities that serve as critical early indicators. First, demographic characteristics, particularly age and biological sex, are pivotal, as they significantly influence symptom expression and diagnostic timing; for instance, females are frequently underdiagnosed or diagnosed later due to a preponderance of internalized symptoms (Young et al., 2020a). Second, physiological comorbidities often co-occur with ADHD, reflecting shared neurobiological underpinnings. Sleep disturbances, ranging from resistance at bedtime to daytime sleepiness, are reported in a substantial proportion of children with ADHD and often exacerbate functional impairments (Faraone et al., 2021; Liang et al., 2023). Similarly, functional elimination disorders, including lower urinary tract symptoms (LUTS) and functional defecation disorders (FDDs), have emerged as significant somatic comorbidities. Our previous work has highlighted the strong association between these elimination issues and ADHD, suggesting they may function as observable somatic warning signs (Liu and Zhang, 2022). Third, emotional and social domains are heavily impacted; emotional dysregulation and peer rejection are pervasive difficulties that frequently precede a formal diagnosis (Paulus et al., 2021). Integrating these multidimensional variables—demographic, physiological, and psychosocial—provides a comprehensive framework for early detection.

Machine learning (ML) offers a promising avenue to integrate these heterogeneous data points into predictive algorithms. Unlike traditional statistical methods, ML can handle high-dimensional interactions between demographics and comorbidities to identify hidden patterns indicative of ADHD risk. While various diagnostic models have been reported, few have focused on a general population-based screening approach that incorporates this specific range of somatic and functional predictors in children.

Therefore, we aimed to develop and validate an interpretable early-warning screening model for ADHD. Distinctively, this model integrates easily assessable indicators from both school (e.g., academic performance) and home settings (e.g., child's sleep, elimination patterns, and parental psychological well-being), reflecting a school-home collaborative perspective. Our ultimate goal is to provide a practical tool that facilitates the timely identification of at-risk children for further professional evaluation.

2. Methods

2.1. Study setting and participants

A cross-sectional survey was carried out from January to December 2023 in a randomly selected city from each of Chongqing, Shaanxi, and Yunnan in China. We used cluster sampling to randomly select twelve primary schools, ten junior high schools, and six senior high schools from a list covering all educational levels. Students aged 6 to 18 years were recruited, resulting in 27,508 returned questionnaires, with a response rate of approximately 75%. The exclusion criteria were as follows: (1) children had an intellectual disability or a confirmed organic disease; (2) children had a history of surgery for the spine, spinal cord, or the urinary or enteric systems; (3) children were suffering from or had a history of urinary tract infections (UTIs).

2.2. Procedure

The survey was conducted using a self-administered, online, anonymous questionnaire completed by the students' primary caregivers. Teachers distributed the questionnaires to the caregivers, and consent was obtained after explaining the study's objectives and risks. Caregivers had the right to refuse participation. To ensure data quality, a rigorous control protocol was implemented. First, prior to data collection, primary caregivers were invited to an online instruction session conducted by a specialized child psychological health physician to provide guidance on understanding the scales, particularly the SNAP-IV. Second, the online system incorporated built-in validation checks that required the completion of all mandatory items and monitored response duration; questionnaires submitted within a timeframe deemed insufficient for serious reading or containing logical inconsistencies were automatically flagged and excluded. The study was approved by the Institutional Review Board of Children's Hospital of Chongqing Medical University (No.2024-73) and was funded as a Chongqing Medical Scientific Research Project, a joint project of the Chongqing Health Commission and Science and Technology Bureau.

2.3. Measures

The comprehensive questionnaire comprised seven sections covering demographics, clinical symptoms, and environmental factors, including data on the child's age, gender, BMI, family characteristics, pregnancy history, and academic performance. The Swanson, Nolan, Pelham-IV (SNAP-IV) scale was used to screen for ADHD risk based on DSM-5 criteria. To comprehensively assess somatic comorbidities, we employed the Children's Sleep Habits Questionnaire (CSHQ) and a specialized urination and defecation questionnaire developed by a pediatrician; the latter evaluated symptom frequency, duration, and onset based on DSM-5 (von Gontard, 2011) and ICCS (Austin et al., 2016) criteria for LUTS, and Rome IV criteria (Hyams et al., 2016) for FDDs. Additional assessments included the Rutter Children's Behavior Questionnaire (RCBQ), the Zung Self-Rating Anxiety/Depression Scales (SAS/SDS) for caregiver mental health, and the Family Environment Scale (FES); detailed psychometric properties and scoring protocols are provided in Supplementary Text S1.

2.4. Data processing and feature engineering

To prepare the raw survey data for machine learning analysis, a standardized processing pipeline was implemented involving data cleaning, scoring transformation, and feature coding. First, regarding data cleaning, we prioritized data authenticity given the large sample size; therefore, a “complete case analysis” approach was adopted where questionnaires containing any missing values in core variables were strictly excluded from the final dataset. Second, regarding scoring transformations, raw item responses were aggregated into total or index scores according to the standard scoring manuals of each instrument. Finally, specific coding strategies were applied to convert categorical variables into numerical formats suitable for the algorithms: nominal variables (e.g., residential area, gender) were processed using One-Hot Encoding to prevent the model from assuming false ordinal relationships, while ordinal variables were mapped to integer sequences to preserve their rank information.

2.5. Model development

The Early Screening Prediction Model for ADHD (ESPM-ADHD) was developed based on the data collected from Chongqing. To ensure methodological rigor and prevent data leakage, the dataset was strictly split into a training set (70%) and an independent internal test set (30%) prior to any feature selection. All model building steps were restricted to the training set. We employed the Least Absolute Shrinkage and Selection Operator (LASSO) regression with 10-fold cross-validation solely on the training data to identify the most robust predictors and eliminate collinear variables. The optimal hyperparameter lambda was

determined to be 0.00083 based on the minimum mean cross-validated error, as illustrated in the cross-validation curve (Fig. S1). Based on the optimal lambda value, 15 key features were selected. Subsequently, to address the class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied using a pipeline approach. This pipeline ensured that synthetic samples were generated only during the training phase dynamically, while the validation data remained authentic. After that, four different machine learning models—XGBoost, Random Forest, Logistic Regression, and LightGBM—were established using Python. Specifically, XGBoost is a prevalent ensemble technique grounded in the gradient boosting framework that amalgamates predictions from multiple weak learners to generate a robust model. Random Forest employs tree-based models, aggregating numerous distinct decision trees through bootstrapping to enhance accuracy and evaluate feature contribution. Binary Logistic Regression serves as a linear baseline, modeling probabilities using a sigmoid function. LightGBM (Light Gradient Boosting Machine) is a distributed gradient boosting framework that uses tree-based learning algorithms. It is designed for higher efficiency and faster training speeds, utilizing a histogram-based algorithm to handle large-scale data effectively.

2.6. Model validation

To evaluate the screening potency of the machine learning models, a held-out validation strategy was employed. The models were trained on the training set (where SMOTE was applied dynamically via a pipeline) and evaluated on the internal test set. Crucially, the internal test set consisted strictly of original, non-oversampled data and was never used during the training or feature selection process. Subsequently, the models were further validated on two external independent datasets from Shaanxi and Yunnan. The modeling process was summarized in Fig. 1.

2.7. Statistical analyses

Statistical analyses were performed using R (v4.2.2) and Python (v3.11.4). Group comparisons were conducted using Welch's *t*-test, Wilcoxon rank-sum test, or Pearson's χ^2/Fisher's exact test, depending on data distribution ($P < .05$). Machine learning models were implemented via Python's scikit-learn, xgboost, and lightgbm packages, utilizing the imbalanced-learn library for SMOTE to address class imbalance. Model performance was evaluated using Sensitivity, Specificity, AUROC, and crucially, AUPRC to ensure predictive reliability in the imbalanced dataset. Optimal probability thresholds for classification were determined using Youden's J statistic (Sensitivity + Specificity −1) to maximize the balance between sensitivity and specificity. Additionally, SHAP analysis was applied to quantify feature contributions, and the optimal model was subsequently deployed as a web-based clinical screening tool.

3. Results

3.1. Characteristics of the study population

A total of 26,726 questionnaires were included in this study according to the inclusion and exclusion criteria, with 554 children at risk of ADHD. The demographic, urination and defecation, family, sleep, behavior characteristics of the 15,085 children (7474 [49.5%] male and 7611 [50.5%] female). The anxiety and depression data of their caregivers were also collected. The baseline characteristics comprising 62

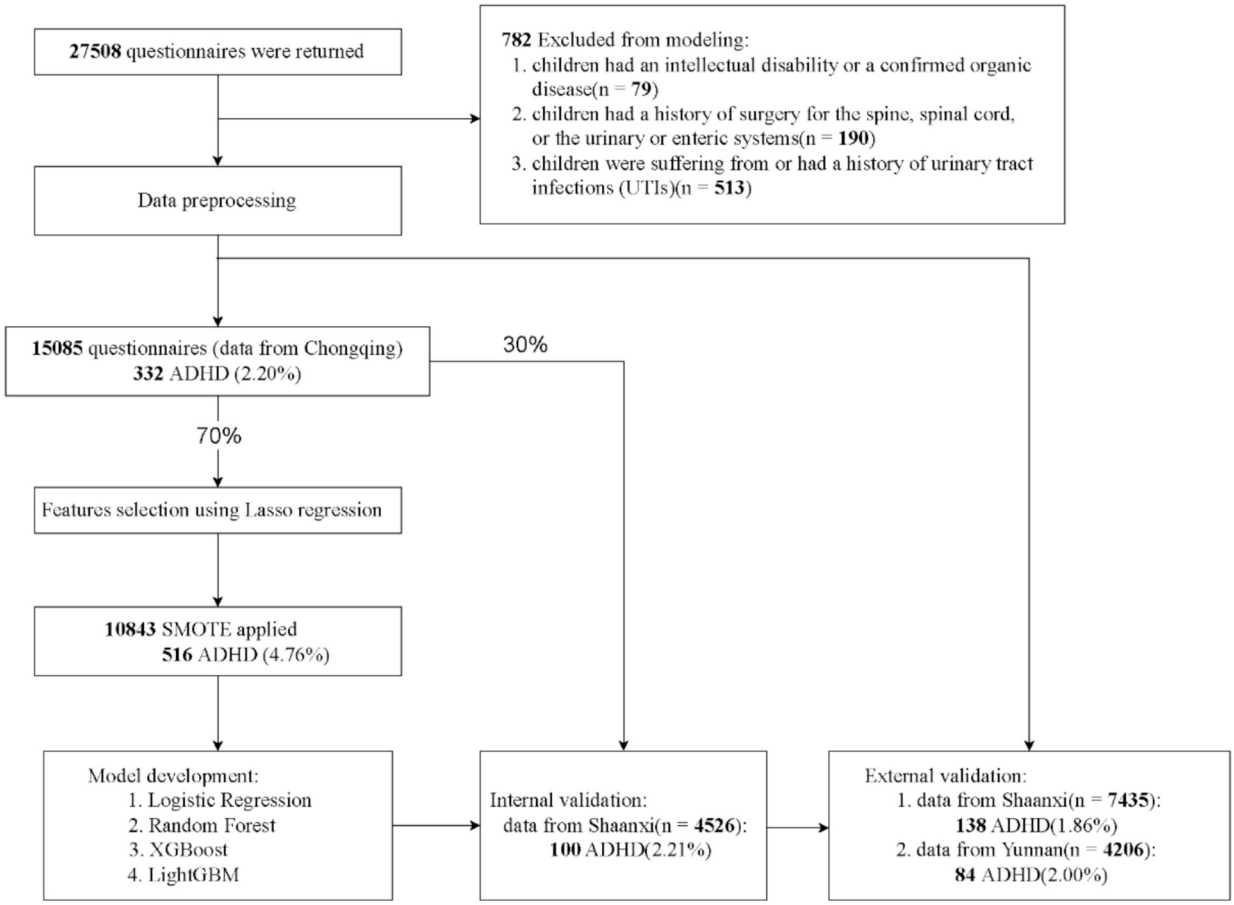


Fig. 1. Overview of Modeling Process. SMOTE indicates synthetic minority oversampling technique.

features are shown in Table 1. A total of 332 children (2.20%) screened positive for ADHD risk. The proportion of male children was higher in the ADHD risk group than the non-ADHD risk group (206 [62.05%] vs 7268 [49.226%]; $P < .001$). Children in the ADHD risk group had poorer Chinese language and math performance, poorer interpersonal relationships, and more severe anxiety, depression symptoms, and suicidal situations compared to those in the non-ADHD risk group ($P < .001$). Significant differences were observed between the groups in terms of all other statistical features derived from Standardized questionnaire about defecation and urination, FES, CSHQ, RCBQ, SDS, and SAS.

3.2. Training and testing data sets

The dataset from Chongqing ($N = 15,085$) served as the basis for model development and was randomly partitioned into a training set (70%, $n = 10,559$) and an independent internal test set (30%, $n = 4526$). Table S1 presents the detailed distributions of the training and internal testing sets, which showed no significant differences. In the training set, to mitigate the impact of class imbalance, SMOTE was applied exclusively to generate synthetic samples for the minority class. This resulted in a more balanced training distribution, consisting of 10,843 participants, with 516 in the ADHD risk group and 10,327 in the non-ADHD risk group.

Crucially, to evaluate model performance in a real-world setting, no data augmentation was applied to the internal test set or the external validation sets. The internal test set retained its original composition with 100 (2.20%) children at risk of ADHD. The external validation sets comprised 7435 children from Shaanxi (138 [1.86%] at risk) and 4206 children from Yunnan (84 [2.00%] at risk). A comparison among the three datasets (Chongqing, Shaanxi, and Yunnan) is presented in Table S2.

3.3. Development of ESPM-ADHD

To strictly prevent data leakage, Lasso regression analysis was conducted exclusively on the training dataset before SMOTE. The deviance residuals and coefficient paths were calculated to identify optimal predictors (Fig. S1). Initially, the LASSO analysis identified 32 potential features with non-zero coefficients. One feature was removed to reduce collinearity (total scores of RCBQ), because antisocial behavior score and neurotic behavior score were part of total score and could provide more detailed information. From the refined set, the top 15 features with the highest absolute coefficients were retained for the final model development. These features included (antisocial behavior score, enuresis, suicidal situations, neurotic behavior score, depressive symptoms, intensity of anxiety symptoms of SAS, Chinese language performance, math performance, excessive volitional stool retention, daytime sleepiness, interpersonal relationships at school, frequency of daytime incontinence, encopresis, anxiety symptoms, and functional constipation). The coefficients of these selected variables in the Lasso regression are detailed in Table S3. Using these 15 key features, four machine learning models (XGBoost classification, Random Forest classification, Binary Logistic Regression, and LightGBM classification) were generated to screen for ADHD risk in children.

3.4. Model performance and validation

In the internal validation, Random Forest classification demonstrated the most robust predictive value among the four machine learning models. It achieved the highest AUPRC of 0.258 and an AUROC of 0.897. At an optimal classification threshold of 0.031 determined by Youden's index, the Random Forest model yielded a Sensitivity of 0.830, Specificity of 0.817, and an F1-Score of 0.168. While Binary Logistic Regression also performed well (AUPRC = 0.233, AUROC = 0.885), LightGBM (AUPRC = 0.199) and XGBoost (AUPRC = 0.166) exhibited comparatively lower performance in identifying positive cases within

Table 1
Characteristics of children in the training dataset.

	Overall[1]	Non-ADHD[1]	ADHD[1]	*P* Value
N	15,085	14,753 (97.80%)	332 (2.20%)	
Demographic characteristics				
Male	7474 (49.55%)	7268 (49.26%)	206 (62.05%)	<0.001[2]
Age	10.00 (8.00, 13.00)	10.00 (8.00, 13.00)	10.00 (8.00, 12.00)	0.050[3]
Height (cm)	145 (130, 155)	145 (130, 155)	140 (130, 152)	0.005[3]
Weight (kg)	40 (28, 50)	40 (28, 50)	38 (29, 49)	0.336[3]
Body Mass Index	17.9 (15.6, 21.3)	17.9 (15.6, 21.3)	18.4 (15.6, 21.6)	0.300[3]
Residential area				0.845[2]
City	7014 (46.50%)	6861 (46.51%)	153 (46.08%)	
City-country fringe	3532 (23.41%)	3450 (23.39%)	82 (24.70%)	
Country	4539 (30.09%)	4442 (30.11%)	97 (29.22%)	
Annual family income (yuan)				0.615[4]
Below 30,000	6104 (40.46%)	5965 (40.43%)	139 (41.87%)	
30,000-80,000	5439 (36.06%)	5319 (36.05%)	120 (36.14%)	
80,000-150,000	2357 (15.62%)	2302 (15.60%)	55 (16.57%)	
150,000-300,000	981 (6.50%)	967 (6.55%)	14 (4.22%)	
300,000-1000,000	170 (1.13%)	166 (1.13%)	4 (1.20%)	
Above 1000,000	34 (0.23%)	34 (0.23%)	0 (0.00%)	
Primary caregiver				0.132[4]
Mother	9512 (63.06%)	9322 (63.19%)	190 (57.23%)	
Father	1216 (8.06%)	1185 (8.03%)	31 (9.34%)	
Paternal grandparents	2674 (17.73%)	2601 (17.63%)	73 (21.99%)	
Maternal grandparents	1041 (6.90%)	1019 (6.91%)	22 (6.63%)	
Nanny	16 (0.11%)	15 (0.10%)	1 (0.30%)	
Others	626 (4.15%)	611 (4.14%)	15 (4.52%)	
Primary caregiver's education				0.236[4]
Junior high and below	9860 (65.36%)	9636 (65.32%)	224 (67.47%)	
Senior high	3257 (21.59%)	3198 (21.68%)	59 (17.77%)	
Bachelor	1907 (12.64%)	1860 (12.61%)	47 (14.16%)	
Master degree and above	61 (0.40%)	59 (0.40%)	2 (0.60%)	
Marital status of parents				0.012[4]
Non-special	11,975 (79.38%)	11,734 (79.54%)	241 (72.59%)	
Divorced	2028 (13.44%)	1972 (13.37%)	56 (16.87%)	
Reconstituted	874 (5.79%)	847 (5.74%)	27 (8.13%)	
Widowed	208 (1.38%)	200 (1.36%)	8 (2.41%)	
Multiple-child family	9244 (61.28%)	9055 (61.38%)	189 (56.93%)	0.100[2]
Gestational weeks				0.172[2]
<32 w	604 (4.00%)	589 (3.99%)	15 (4.52%)	
32–35 w	775 (5.14%)	749 (5.08%)	26 (7.83%)	
35–37 w	1256 (8.33%)	1234 (8.36%)	22 (6.63%)	
Term (37–40 w)	11,452 (75.92%)	11,206 (75.96%)	246 (74.10%)	
>40 w	998 (6.62%)	975 (6.61%)	23 (6.93%)	
Birth weight				0.238[4]
<1 kg	71 (0.47%)	67 (0.45%)	4 (1.20%)	
1–1.5 kg	528 (3.50%)	513 (3.48%)	15 (4.52%)	

(*continued on next page*)

Table 1 (*continued*)

	Overall[1]	Non-ADHD[1]	ADHD[1]	*P* Value
1.5–2.5 kg	2068 (13.71%)	2023 (13.71%)	45 (13.55%)	
2.5–4 kg	11,844 (78.52%)	11,590 (78.56%)	254 (76.51%)	
>4 kg	574 (3.81%)	560 (3.80%)	14 (4.22%)	
Smoking in pregnancy	409 (2.71%)	400 (2.71%)	9 (2.71%)	>0.999[2]
Alcohol drinking in pregnancy	217 (1.44%)	211 (1.43%)	6 (1.81%)	0.484[4]
Passive smoking in pregnancy	2298 (15.23%)	2207 (14.96%)	91 (27.41%)	<0.001[2]
Depressed in pregnancy				
Mother	1176 (7.80%)	1099 (7.45%)	77 (23.19%)	<0.001[2]
Father	233 (1.54%)	218 (1.48%)	15 (4.52%)	<0.001[2]
Chinese language performance				<0.001[2]
Top few	1508 (10.00%)	1499 (10.16%)	9 (2.71%)	
Above average	4822 (31.97%)	4789 (32.46%)	33 (9.94%)	
Average	6143 (40.72%)	6045 (40.97%)	98 (29.52%)	
Below average	2612 (17.32%)	2420 (16.40%)	192 (57.83%)	
Math performance				<0.001[2]
Top few	1828 (12.12%)	1815 (12.30%)	13 (3.92%)	
Above average	4820 (31.95%)	4782 (32.41%)	38 (11.45%)	
Average	5746 (38.09%)	5633 (38.18%)	113 (34.04%)	
Below average	2691 (17.84%)	2523 (17.10%)	168 (50.60%)	
Personality				0.761[2]
Introverted	6828 (45.26%)	6675 (45.25%)	153 (46.08%)	
Extroverted	8257 (54.74%)	8078 (54.75%)	179 (53.92%)	
Interpersonal relationships at school				<0.001[4]
Good	8422 (55.83%)	8342 (56.54%)	80 (24.10%)	
Average	6510 (43.16%)	6289 (42.63%)	221 (66.57%)	
Poor	153 (1.01%)	122 (0.83%)	31 (9.34%)	
Depressive symptoms				<0.001[4]
Not at all	10,104 (66.98%)	9981 (67.65%)	123 (37.05%)	
A little	4585 (30.39%)	4434 (30.05%)	151 (45.48%)	
Quite a bit	279 (1.85%)	248 (1.68%)	31 (9.34%)	
Very much	117 (0.78%)	90 (0.61%)	27 (8.13%)	
Anxiety symptoms				<0.001[4]
Not at all	9711 (64.38%)	9591 (65.01%)	120 (36.14%)	
A little	4902 (32.50%)	4761 (32.27%)	141 (42.47%)	
Quite a bit	328 (2.17%)	288 (1.95%)	40 (12.05%)	
Very much	144 (0.95%)	113 (0.77%)	31 (9.34%)	
Suicidal situations				<0.001[4]
None	14,785 (98.01%)	14,484 (98.18%)	301 (90.66%)	
Suicidal thoughts	263 (1.74%)	241 (1.63%)	22 (6.63%)	
Suicidal behavior	37 (0.25%)	28 (0.19%)	9 (2.71%)	
Standardized questionnaire about defecation and urination				
Enuresis	939 (6.22%)	871 (5.90%)	68 (20.48%)	<0.001[2]
Frequency of enuresis				<0.001[4]
None	14,538 (96.37%)	14,245 (96.56%)	293 (88.25%)	
Once a month	278 (1.84%)	263 (1.78%)	15 (4.52%)	
Twice a month	81 (0.54%)	74 (0.50%)	7 (2.11%)	
3–4 times a month	69 (0.46%)	61 (0.41%)	8 (2.41%)	
5–6 times a month	69 (0.46%)	64 (0.43%)	5 (1.51%)	
Almost every day	50 (0.33%)	46 (0.31%)	4 (1.20%)	
Daytime incontinence	797 (5.28%)	755 (5.12%)	42 (12.65%)	<0.001[2]
Frequency of daytime incontinence				<0.001[2]
None	14,084 (93.36%)	13,812 (93.62%)	272 (81.93%)	
<1/week	591 (3.92%)	561 (3.80%)	30 (9.04%)	
1/week	134 (0.89%)	128 (0.87%)	6 (1.81%)	
2–3/week	166 (1.10%)	152 (1.03%)	14 (4.22%)	
4–5/week	57 (0.38%)	53 (0.36%)	4 (1.20%)	
Almost every time around urination	53 (0.35%)	47 (0.32%)	6 (1.81%)	
Frequency of urination				0.023[2]
≤2/d	2802 (18.57%)	2738 (18.56%)	64 (19.28%)	
3–7/d	11,995 (79.52%)	11,740 (79.58%)	255 (76.81%)	
≥8/d	288 (1.91%)	275 (1.86%)	13 (3.92%)	
Urinary urgency	993 (6.58%)	930 (6.30%)	63 (18.98%)	<0.001[2]
Holding maneuvers	962 (6.38%)	901 (6.11%)	61 (18.37%)	<0.001[2]
Urinary hesitancy	1013 (6.72%)	972 (6.59%)	41 (12.35%)	<0.001[2]
Intermittency	452 (3.00%)	416 (2.82%)	36 (10.84%)	<0.001[2]
Functional constipation	779 (5.16%)	727 (4.93%)	52 (15.66%)	<0.001[2]
Excessive volitional stool retention	614 (4.07%)	578 (3.92%)	36 (10.84%)	<0.001[2]
Encopresis	1147 (7.60%)	1059 (7.18%)	88 (26.51%)	<0.001[2]
Treatment of enuresis				<0.001[4]
Untreated	470 (3.12%)	437 (2.96%)	33 (9.94%)	
Treated	77 (0.51%)	71 (0.48%)	6 (1.81%)	
Excerpt from the Family Environment Scale (FES)				
Cohesion	10 (4, 14)	10 (4, 14)	3 (0, 9)	<0.001[3]
Conflict	−7.0 (−11.0, −4.0)	−7.0 (−11.0, −4.0)	−3.0 (−7.0, 0.0)	<0.001[3]
The Children's Sleep Habits Questionnaire (CSHQ)				
Bedtime resistance	7.00 (6.00, 9.00)	7.00 (6.00, 9.00)	8.00 (7.00, 11.00)	<0.001[3]
Sleep onset delay	1.00 (1.00, 2.00)	1.00 (1.00, 2.00)	1.00 (1.00, 2.00)	<0.001[3]
Sleep duration	5.00 (3.00, 6.00)	5.00 (3.00, 6.00)	5.00 (4.00, 7.00)	<0.001[3]
Sleep anxiety	4.00 (4.00, 6.00)	4.00 (4.00, 6.00)	5.00 (4.00, 7.00)	<0.001[3]
Night wakings	3.00 (3.00, 3.00)	3.00 (3.00, 3.00)	3.00 (3.00, 4.00)	<0.001[3]
Parasomnias	7.00 (7.00, 8.00)	7.00 (7.00, 8.00)	8.00 (7.00, 10.00)	<0.001[3]
Sleep disordered breathing	3.00 (3.00, 3.00)	3.00 (3.00, 3.00)	3.00 (3.00, 4.00)	<0.001[3]
Daytime sleepiness	12.00 (10.00, 14.00)	12.00 (10.00, 14.00)	14.00 (12.00, 16.00)	<0.001[3]
Total score	42.00 (38.00, 46.00)	42.00 (38.00, 46.00)	48.00 (43.00, 54.00)	<0.001[3]
Sleep disorders	7807 (51.75%)	7534 (51.07%)	273 (82.23%)	<0.001[2]
The parent-reported Rutter Children's Behavior Questionnaire (RCBQ)				
Antisocial behavior (A behavior) score	1.00 (0.00, 2.00)	1.00 (0.00, 2.00)	3.00 (2.00, 5.00)	<0.001[3]

(*continued on next page*)

Table 1 (*continued*)

	Overall[1]	Non-ADHD[1]	ADHD[1]	*P* Value
Neurotic behavior (N behavior) score	1.00 (0.00, 2.00)	1.00 (0.00, 2.00)	2.00 (1.00, 4.00)	<0.001[3]
Total score	5.00 (1.00, 11.00)	5.00 (1.00, 10.00)	18.00 (12.00, 26.00)	<0.001[3]
Abnormal behavior	2593 (17.19%)	2363 (16.02%)	230 (69.28%)	<0.001[2]
Type of abnormal behavior				<0.001[2]
Antisocial behavior	1035 (6.86%)	903 (6.12%)	132 (39.76%)	
Neurotic behavior	981 (6.50%)	919 (6.23%)	62 (18.67%)	
Mixed behavior	577 (3.82%)	541 (3.67%)	36 (10.84%)	
The Zung Self-Rating Depression Scale (SDS) in caregivers				
Total score	45.00 (36.00, 56.00)	45.00 (36.00, 56.00)	53.00 (44.00, 61.00)	<0.001[3]
Depression symptoms	6009 (39.83%)	5806 (39.35%)	203 (61.14%)	<0.001[2]
Intensity of depression symptoms				<0.001[4]
Mild	3004 (19.91%)	2895 (19.62%)	109 (32.83%)	
Moderate	2093 (13.87%)	2034 (13.79%)	59 (17.77%)	
Severe	47 (0.31%)	42 (0.28%)	5 (1.51%)	
The Zung Self-Rating Anxiety Scale (SAS) in caregivers				
Total score	40.00 (34.00, 44.00)	40.00 (34.00, 44.00)	45.00 (40.00, 51.00)	<0.001[3]
Anxiety symptoms	1282 (8.50%)	1178 (7.98%)	104 (31.33%)	<0.001[2]
Intensity of anxiety symptoms				<0.001[4]
Mild	1068 (7.08%)	1000 (6.78%)	68 (20.48%)	
Moderate	170 (1.13%)	139 (0.94%)	31 (9.34%)	
Severe	44 (0.29%)	39 (0.26%)	5 (1.51%)	

Abbreviations: ADHD, Attention-deficit/hyperactivity disorder.
[1] n (%); Median (IQR).
[2] Pearson's Chi-squared test.
[3] Wilcoxon rank sum test.
[4] Fisher's exact test.

this imbalanced setting. Fig. S2 displays the discrimination and calibration of the model in the internal validation dataset via precision-recall and calibration curves.

The generalization capability of the models was further assessed using external datasets from Shaanxi and Yunnan. As shown in Fig. 2 and Table S4, all models maintained high discrimination on unseen data. In the Shaanxi dataset, both Logistic Regression (AUPRC = 0.347, AUROC = 0.932) and Random Forest (AUPRC = 0.298, AUROC = 0.929) showed exceptional performance. Specifically, using the optimal threshold of 0.026 derived from Youden's J statistic, the Random Forest model achieved a remarkable Sensitivity of 0.906, alongside a Specificity of 0.833 and an Accuracy of 0.834. Similarly, in the Yunnan dataset, Random Forest outperformed the other models, achieving the highest AUPRC (0.253) and AUROC (0.907). At a threshold of 0.038, it maintained a robust Sensitivity of 0.821 and a Specificity of 0.858.

Although Logistic Regression showed slightly higher metrics in the Shaanxi cohort, Random Forest demonstrated the most consistent stability across all three datasets (Internal, Shaanxi, and Yunnan), particularly in maintaining a balanced trade-off between sensitivity and specificity. Consequently, Random Forest classification was deemed the best-performing model for the final construction of the screening tool.

3.5. Feature importance

The SHAP algorithm was used to obtain the importance of each feature to the outcome predicted by the models. Feature importance plots for the four models were shown in Fig. 3. For the Random Forest model, the top 10 features contributing to the screening accuracy (listed in descending order of importance) were: antisocial behavior score, Chinese language performance, daytime sleepiness, math performance, neurotic behavior score, anxiety symptoms, interpersonal relationships at school, depressive symptoms, encopresis, intensity of anxiety symptoms of SAS. Antisocial behavior score, Chinese language performance and daytime sleepiness were included among the 4 most important features in all models (Fig. 3).

3.6. Explanation of the model constructed by Random Forest

As shown in Fig. 4A, the horizontal axis represents the impact of each feature on the prediction value, indicating whether it is associated with a higher or lower prediction. For each feature, high and positive values are depicted in red, while low and negative values are shown in blue. Fig. 4B and C showed the individual force plots for children with (B) ADHD risk and (C) non-ADHD risk. The SHAP values highlighted the predictive features for each child and the contribution of each feature to ADHD risk. Red features indicated an increased risk of ADHD, while blue features indicated a decreased risk. The magnitude of each feature's effect was visualized by the length of the arrows, with longer arrows representing a greater impact on the outcome.

3.7. Development of an online web-calculator based on Random Forest

To facilitate the translation of our research findings into clinical practice, we developed a user-friendly online web calculator based on the Random Forest model. Although the Random Forest algorithm involves complex ensemble learning processes that are difficult to calculate manually, it demonstrated the most robust and stable predictive performance across our multi-center datasets. The web-based tool allows clinicians and caregivers to input the child's specific data regarding the 15 identified risk features, and instantly estimates the probability of the child being at risk for ADHD. As illustrated in Fig. S3, the calculator not only provides risk stratification but also visualizes individual feature contributions via SHAP plots to enhance clinical interpretability. This tool incorporates the exact model parameters and thresholds validated in this study and is freely accessible at: https://kzltphi2wstrpqewwyabtb.streamlit.app/.

4. Discussion

Our study developed and validated an explainable, machine learning-based Early Screening Prediction Model (ESPM-ADHD) for identifying children at risk of ADHD in a large-scale, multi-center Chinese population. By integrating demographic characteristics, academic performance, and, crucially, somatic comorbidities (sleep and elimination disorders), our Random Forest model achieved robust screening performance across internal and external validation cohorts. These findings suggest that a multidimensional approach, leveraging readily available data from parents and schools, can effectively triage children for further clinical evaluation, addressing the critical bottleneck in early ADHD identification. Importantly, the optimal probability thresholds identified in this study (determined via Youden's J statistic) serve as clinically actionable referral cut-offs. Specifically, any student with a predicted risk score exceeding this threshold is flagged for priority professional assessment. This strategy ensures high sensitivity to minimize missed diagnoses, validating the tool's utility as a reliable first-step filter in school-based screening scenarios.

Consistent with previous research, our SHAP analysis identified behavioral dysregulation (specifically antisocial and neurotic behaviors)

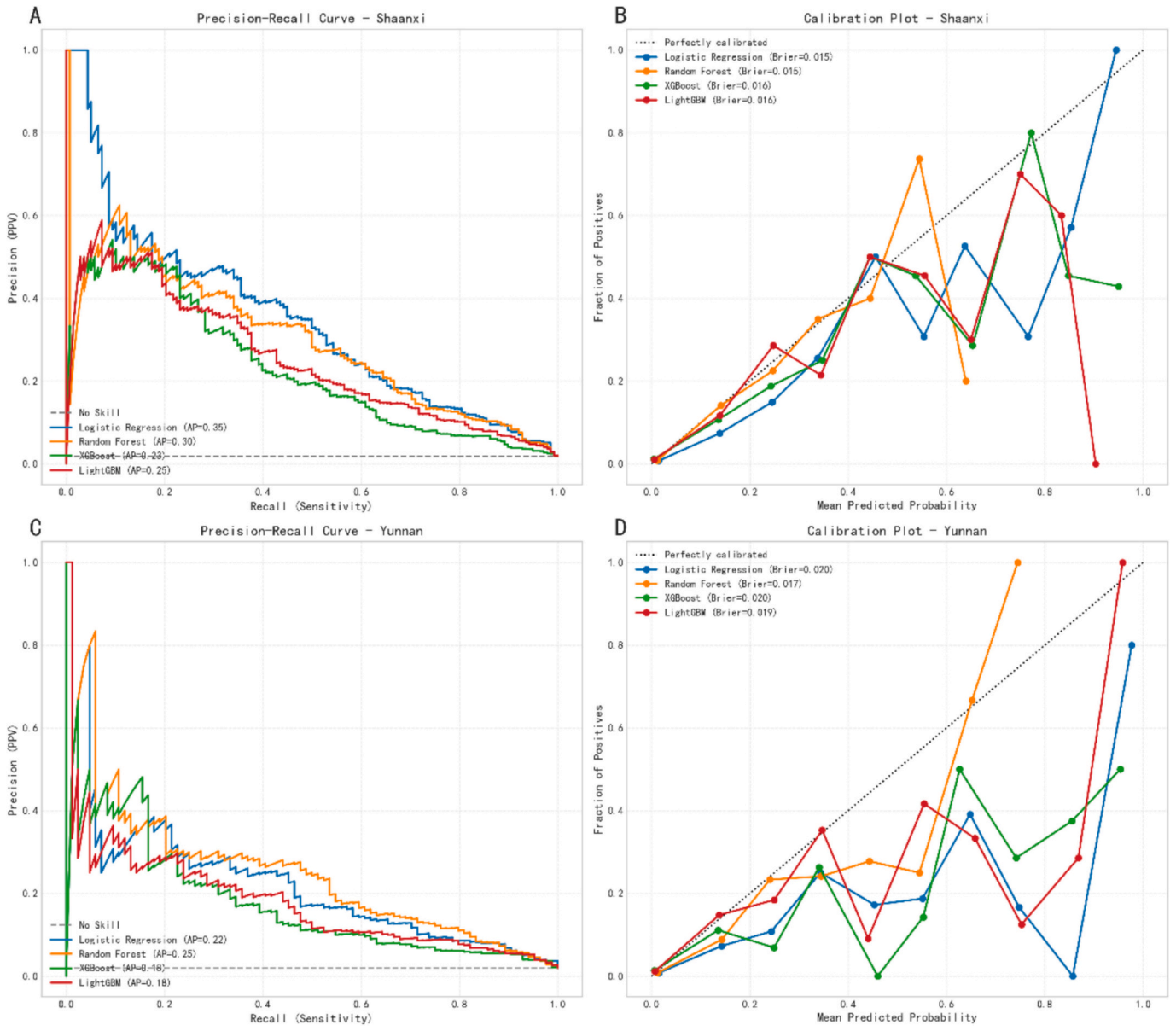


Fig. 2. Performance of machine learning models in external validation cohorts. (A, C) Precision-Recall Curves (PRC) illustrating the discriminative performance of the four models in Shaanxi (A) and Yunnan (C). The gray dashed line represents the baseline performance (no-skill classifier). (B, D) Calibration plots demonstrating the agreement between predicted probabilities and observed frequencies in Shaanxi (B) and Yunnan (D).
Abbreviations: AP, average precision; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.

and academic underachievement (Chinese and Math performance) as the top predictors for ADHD risk. This finding aligns with the contemporary conceptualization of ADHD, which extends beyond simple attention deficits to a broader impairment in executive functioning, fundamentally affecting self-regulation and goal-directed behavior (Champ et al., 2023). The strong predictive value of academic difficulties in our model highlights the functional cascade often observed in these children; academic underachievement is frequently one of the earliest and most persistent functional impairments, often emerging prior to a formal clinical diagnosis (Arnold et al., 2020). Furthermore, the prominence of "antisocial behavior" as a leading predictor echoes established evidence regarding care pathways: children with externalizing, disruptive behaviors are significantly more likely to be recognized and referred for clinical evaluation compared to those with purely internalizing symptoms (Sayal et al., 2018). In cultural contexts where academic success is heavily emphasized, the intersection of behavioral disruption and scholastic failure serves as a critical trigger for parental help-seeking (Pang et al., 2021). By quantitatively capturing these specific functional impairments, our model serves as an objective "red flag" system, enabling primary care providers to identify high-risk children who might otherwise be overlooked due to time constraints or the subtlety of symptom presentation.

A distinguishing feature of our study is the integration of somatic comorbidities—specifically sleep disturbances and elimination disorders—into the screening algorithm, shifting the focus from purely behavioral symptoms to a broader physiological phenotype. The identification of daytime sleepiness as a top-tier predictor corroborates the growing consensus that circadian rhythm disruption is not merely a comorbid artifact but a core pathophysiological component of ADHD, potentially driven by delayed circadian phase and altered melatonin profiles that exacerbate neurocognitive deficits (Becker, 2020). Furthermore, the inclusion of elimination disorders (enuresis, encopresis, and functional constipation) represents a critical advancement, leveraging the "bladder-gut-brain axis" for early detection. These

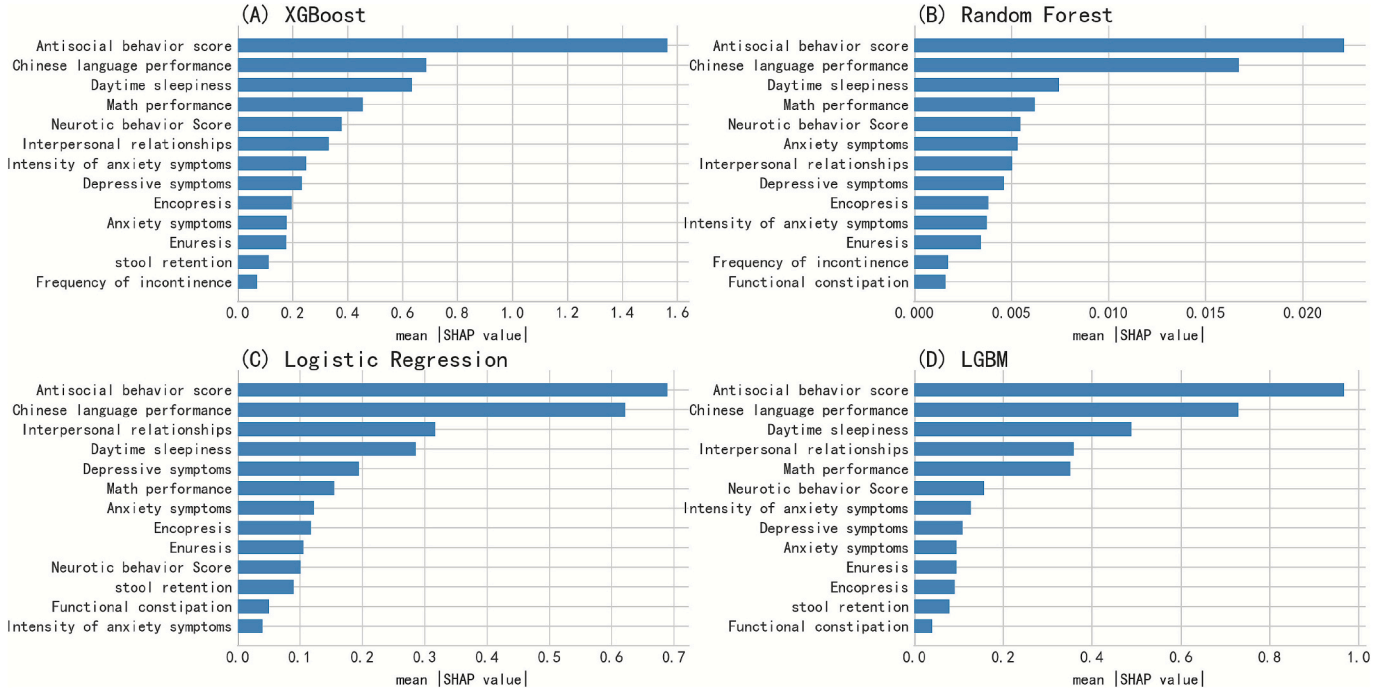


Fig. 3. Feature Importance Plots for the four models. The importance ranking of the thirteen features in four models according to the mean |SHAP value|, SHAP indicates the Shapley additive explanation machine learning algorithm. (A) for XGBoost classification, (B) for random forest classification, (C) for binary logistic regression and (D) for LightGBM classification.

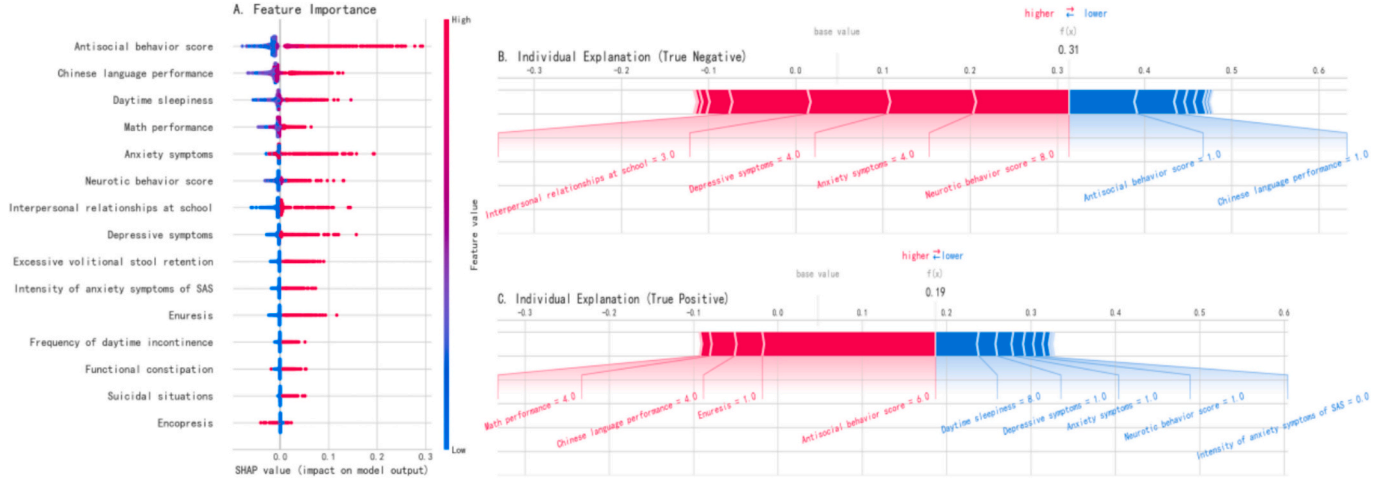


Fig. 4. The interpretation of significant features of the model constructed by random forest classification. (A) The importance ranking of features based on mean absolute SHAP values. (B) Individual-level interpretation for a child predicted to be at low risk. (C) Individual-level interpretation for a child predicted to be at high risk for ADHD.

somatic symptoms are frequently overlooked in psychiatric screenings despite sharing underlying neurobiological mechanisms with ADHD, particularly the delayed maturation of central inhibitory pathways and dopaminergic dysregulation in the prefrontal cortex and pontine micturition center (von Gontard et al., 2022; von Gontard and Equit, 2015). By incorporating these markers, our model effectively captures a "somatic subtype" of neurodevelopmental vulnerability, minimizing missed cases among children who may present with physiological dysregulation rather than overt disruptive behavior.

Methodologically, the superior stability of the Random Forest model across diverse geographical cohorts (Chongqing, Shaanxi, and Yunnan) underscores the critical advantage of non-linear machine learning algorithms over traditional logistic regression in handling complex biopsychosocial data. Unlike linear models that rely on additive assumptions, tree-based ensemble methods effectively capture high-order, non-linear interactions between heterogeneous features—such as the synergistic risk amplification between physiological comorbidities and psychosocial stressors—thereby enhancing predictive generalizability in real-world populations (Shatte et al., 2019). However, algorithmic sophistication alone does not guarantee clinical utility. To bridge the persistent "translational gap" between computational innovation and bedside practice, we deployed the ESPM-ADHD as an accessible web-based calculator. This deployment transforms a complex algorithm into a pragmatic, probability-based decision support tool, empowering pediatricians to make evidence-based referral decisions. This approach directly aligns with the emerging paradigm of "precision psychiatry,"

which advocates for the use of data-driven, individualized prediction tools to optimize resource allocation and enable early, targeted intervention (Chekroud et al., 2021).

It is crucial to reiterate that the ESPM-ADHD is a screening and triage tool, not a diagnostic instrument. Its intended use is in community and primary care settings—by school health professionals or pediatricians during well-child visits—to efficiently identify children who warrant a comprehensive, gold-standard diagnostic assessment by a specialist. The developed online calculator lowers the barrier to conducting this initial risk stratification. We envision it initiating a closed-loop management pathway: concerning observations → quick online screening → elevated risk score → structured referral → definitive diagnosis and intervention. This pathway directly addresses the critical delay often experienced between symptom emergence and service access.

5. Limitations

Our study has several limitations. First, the "ADHD risk" classification was based on the SNAP-IV rating scale rather than a gold-standard clinical diagnosis based on semi-structured interviews. While SNAP-IV is a validated and widely used screening tool, false positives are possible. Consequently, our model should be viewed as a risk stratification tool rather than a diagnostic instrument. Second, the data were collected via parent-reported questionnaires, which may introduce reporting bias; however, the large sample size helps mitigate individual variance. Third, regarding feature generalizability, the inclusion of "Chinese language performance" reflects the specific educational context of our study population, where this subject represents a core metric of literacy and executive function. While this specific label limits direct application to non-Chinese speaking populations, this feature essentially serves as a proxy for "native language academic performance." Future adaptations of this model in other countries should substitute this feature with the corresponding local native language metric to capture the universal impact of ADHD on literacy-related academic achievement. Finally, while SMOTE effectively balanced the training process, the extreme class imbalance in real-world settings means that the Positive Predictive Value (PPV) remains modest, emphasizing the need for clinical confirmation of positive screening results.

6. Conclusions

We have developed and validated a practical screening tool (ESPM-ADHD) that leverages multidimensional, easily observable features from both school and home environments. This school-home integrated approach offers a novel and pragmatic strategy for the early detection of children at risk for ADHD in community settings. By providing an accessible online calculator, our work aims to bridge the gap between initial concerns and professional evaluation, ultimately promoting earlier intervention and improving outcomes for children with ADHD.

CRediT authorship contribution statement

Chunbo Wang: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Zhijuan Li:** Resources, Investigation, Data curation, Conceptualization. **Jiangling Su:** Resources, Investigation, Data curation, Conceptualization. **Mo Wang:** Writing – review & editing, Supervision, Methodology. **Gaofu Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2026.121247.

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